

Matias C.Y. Relyea

CONTACT INFORMATION	<i>E-mail:</i> hyanlong@unc.edu <i>Website:</i> matiascyrelyea.github.io
EDUCATION	University of North Carolina at Chapel Hill, Chapel Hill, North Carolina B.S. Mathematics; Minor in violin performance (expected graduation, May 2029). <i>Relevant Coursework:</i> Undergraduate: Differential Geometry; Topology (Duke University); Algebraic Geometry; Algebraic Structures; Advanced Calculus. Graduate: Modules, Linear Algebra, and Groups.
PUBLICATIONS	[1] Matias C.Y. Relyea. The Quadratic and Cubic Characters of 2. <i>Mathematics Magazine</i>. Published online January 2026. [arXiv: 2410.21646] [DOI: 10.1080/0025570X.2025.2574223].
PREPRINTS	[2] Harin Jung, Aiden Novick, Matias C.Y. Relyea. Improved Poly-Log Diameter Bounds for $SL_2(\mathbb{Z}_p)$ via Higher Commutators. <i>In preparation</i>. [3] Matias C.Y. Relyea. On Finite Fields and Higher Reciprocity (2024). [arXiv: 2407.03559].
PRESENTATIONS	[4] “Reciprocity Laws” (chalkboard lecture series). Carolina Math Club, Chapel Hill, North Carolina. March and April 2026. [5] “On Finite Fields and Higher Reciprocity” (poster). NCSSM Research Symposium Spring 2024.
RESEARCH EXPERIENCE	Einstein Institute of Mathematics, Hebrew University of Jerusalem (HUJI) REU Visiting Research Fellow, Summer 2026 - Extended the Solovay-Kitaev theorem for compact Lie groups, proving new diameter bounds for $SL_2(\mathbb{Z}/p^n\mathbb{Z})$ using Elkasapy words and higher commutators. Quantum computation, Lie theory, group theory, and algebraic number theory. Mentored by Professor Shai Evra. <i>Manuscript in preparation</i>. University of North Carolina at Chapel Hill, Chapel Hill, North Carolina Undergraduate Researcher, Fall 2026—Present - Ongoing research in birational geometry, Lie theory, and group theory. Mentored by Professor Olivia Dumitrescu. Directed Reading Program on Simplicial Homology and Cohomology, Fall 2026 Directed Reading Program on Category Theory, Fall 2025

North Carolina School of Science and Mathematics (NCSSM), Durham, North Carolina

Independent studies on number theory and algebraic number theory, producing one arXiv preprint and one peer-reviewed publication. August 2023—May 2025. Mentored by Professor Tamar Avineri.

Euler Circle, Mountain View, California

Attended Summer 2022 Independent Research and Paper Writing course. Independently produced 44-page expository paper on quadratic reciprocity. Mentored by Professor Simon Rubinstein-Salzedo.

SEMINARS AND CONFERENCES ATTENDED [6] 25th Midrasa Mathematicae: Groups, Expanders, and Codes. Israel Institute for Advanced Studies (IAS), 14-18 June, 2026.

OTHER WRITING [7] Peyton Jackson, Tatiana Medved, and Matias C.Y. Relyea. “Can You Even?” An Exploration of Polyominoes. *NCSSM Broad Street Scientific, 2024-2025* ([link](#)). Published online May 2026. [[PDF](#)].

[8] Matias C.Y. Relyea. Proofs and Applications of Quadratic Reciprocity. Summer 2022, *Euler Circle*. [[PDF](#)].

TEACHING AND SERVICE **Mathematics Institute of the Triangle, Chapel Hill, North Carolina**

Mathematics Tutor and Teaching Assistant, Fall 2025—Present

- Provided one-on-one tutoring to students located in the Chapel Hill area. Assisted with grading and led short problem-solving workshops.

University of North Carolina at Chapel Hill, Chapel Hill, North Carolina

Undergraduate Learning Assistant

- MATH 381 (Discrete Mathematics), Fall 2025.

Chapel Hill Math Circle, Chapel Hill, North Carolina

Program Volunteer, Fall 2025—Present

- Collaborated with UNC graduate students to assist and lead teaching for various levels of mathematics circle coursework.

TECHNOLOGICAL SKILLS Proficient in $\text{\LaTeX}$, familiar with Python.